

HTTP Server

Serve a DuckDB connection through HTTP.



Install & load

```
INSTALL httpserver FROM community;
LOAD httpserver;
```

01 At a glance

Start an HTTP listener inside a DuckDB process and query it from an HTTP client. Send SQL in a POST body, receive JSON results, and keep the process running for the lifetime of the service.

HTTP client → SQL request → DuckDB result

02 Server calls & HTTP contract

`httpserve_start(host, port, auth)`

Start the in-process listener with one shared credential.

`httpserve_stop()`

Stop the listener; call start again to change settings.

POST /

Put SQL in the request body to execute it.

GET /?q=...

Pass URL-encoded SQL as a query parameter.

GET /ping

Unauthenticated liveness endpoint.

`default_format=JSONEachRow`

Return newline-delimited JSON objects, one per result row.

`default_format=JSONCompact`

Return metadata, row arrays and query statistics.

03 Keep in mind

Choose the authentication mode

An auth value containing user:pass selects HTTP Basic; a shared key uses X-API-Key. An empty string disables authentication. The listener speaks HTTP: use a TLS proxy for access beyond a trusted local environment.

04 Start a local listener

MANUAL SETUP

```
SELECT httpserve_start(
  'localhost', 9999, 'replace-this-key'
);
```

Run in a dedicated DuckDB process and leave it open. Replace the example key.

05 Read from another DuckDB

REQUIRES SETUP

```
INSTALL httpfs;
LOAD httpfs;
CREATE SECRET quickref_http (
  TYPE HTTP,
  SCOPE 'http://localhost:9999/',
  EXTRA_HTTP_HEADERS MAP {
    'X-API-Key': 'replace-this-key'
  }
);
SELECT * FROM read_json_auto(
  'http://localhost:9999/?q=SELECT%2042'
  || '&default_format=JSONEachRow'
```

Use a second process after the listener starts; the matching key is required.

06 Stop from the serving process

MANUAL SETUP

```
SELECT httpserve_stop();
```

The listener also stops when its DuckDB process exits.

07 In practice

Scope the DuckDB process

Clients can submit SQL using the served connection's capabilities. Bind to localhost for development; configure DuckDB permissions and filesystem access for deployment. Read-only database access alone is not a complete SQL sandbox. Keep the process alive.

[Full docs & examples ↗](#) [Source ↗](#)